

# Fig5 prototype: strict-path syntenic/recombination schematic

Geometry comes from strict primary-path selected\_segments.tsv rows; coordinates are native 500 kb assembly-window offsets, not CHM13 projections.

**Flow legend**  PAR1 positive-control flow  primary PHR donor flow  side fragment  low-confidence tail  same-chromosome context

## 1. PAR1 positive control

PAN027 paternal hap2 <- PAN011 father  
PAR 151 kb; same-chr 349 kb

**source: chrX PAR1 context**

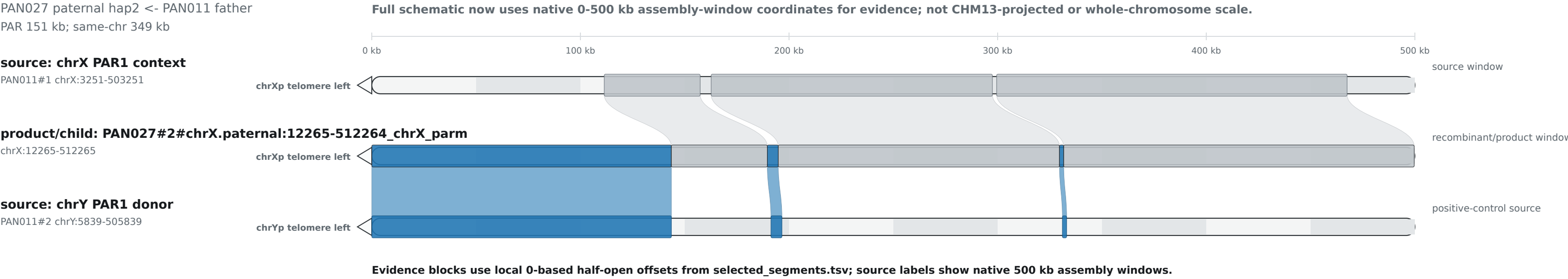
PAN011#1 chrX:3251-503251

**product/child: PAN027#2#chrX.paternal:12265-512264\_chrX\_parm**

chrX:12265-512265

**source: chrY PAR1 donor**

PAN011#2 chrY:5839-505839



## 2. PHR candidate 1: PAN027 chr9q with chr3q donor

PAN027 paternal hap2 <- PAN011 father  
primary donor 45 kb; side fragments 8 kb; low-conf. tail 493 bp; same-chr 446 kb

**source: chr9q same-chromosome context**

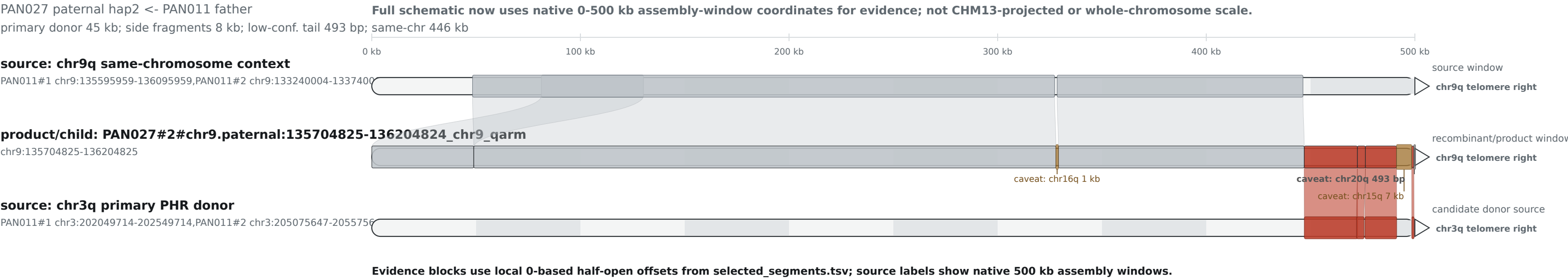
PAN011#1 chr9:135595959-136095959,PAN011#2 chr9:133240004-1337400

**product/child: PAN027#2#chr9.paternal:135704825-136204824\_chr9\_qarm**

chr9:135704825-136204825

**source: chr3q primary PHR donor**

PAN011#1 chr3:202049714-202549714,PAN011#2 chr3:205075647-2055756



## 3. PHR candidate 2: PAN028 strict chr9q path

PAN028 maternal hap1 <- PAN027 mother  
primary donor 34 kb; side fragments 16 kb; same-chr 449 kb

**source: chr9q same-chromosome context**

PAN027#1 chr9:128564684-129064684,PAN027#2 chr9:135704825-1362048

**product/child: PAN028#1#chr9.haplotype1:134380985-134880984\_chr9\_qarm**

chr9:134380985-134880985

**source: chr3q primary PHR donor**

PAN027#1 chr3:200905557-201405557,PAN027#2 chr3:202048279-2025482

